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Intrusion detection model using gene expression programming to optimize parameters of convolutional neural network for energy internet

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A B S T R A C T

The open, interconnected, and shared operational characteristics of the energy Internet introduce more sophisticated cybersecurity attacks. How to accurately detect these cyber attacks is crucial for energy Internet security protection. Existing machine learning-based intrusion detection algorithms cannot cope with the continuous increase of network traffic and features in the energy Internet. And convolutional neural networks (CNN) can be a good solution for the descending and optimal selection of high-dimensional intrusion features. Unfortunately, traditional convolutional neural networks have complex structures with many parameters and are prone to fall into local optimality. To fill the gap of CNN, in this paper, we use a gene expression programming (GEP) to optimize the parameters of CNN and propose an intrusion detection algorithm based on GEP-CNN (GCNN-IDS). Our key idea is to avoid the convolutional neural network from falling into local optimum by designing a new code on GEP and fitness function to optimize the parameters of the CNN using the global search capability of GEP. The experimental results on two benchmark datasets and a real dataset substantiate that the detection accuracy of the optimized CNN-based intrusion detection algorithm (ICNN-IDS) reaches up to 0.9143 under different parameter combinations; meanwhile, compared with other algorithms, the detection accuracy, precision, recall, F_1 and false detection rate of the intrusion detection model proposed in this paper reach 0.9897, 0.99, 0.98, 0.97 and 0.0126, respectively.

1. Introduction

The large-scale development and utilization of renewable energy will become the development trend in the global energy field, and it also faces the problems of how to realize the widespread interconnection of distributed renewable energy and improve the efficiency and flexibility of various energy systems. Energy Internet based on the Internet concept has become the main development direction of the energy industry [1].

The in-depth application of 5G and edge computing in the energy Internet makes the existing security defense boundary of the energy Internet migrate to the user side. At the same time, industrial control systems in power and oil sectors are changing from closed structures to open structures, and their production environments are changing from manual or semi-automated forms to networked automation forms. More and more terminal devices for collection, monitoring and control are being built, managed and controlled through various wired or wireless public network. The energy interaction, communication and information interaction among various types of energy sources in the Energy Internet are mostly realized by industrial control systems. With the widespread application of IoT, edge computing, blockchain and artificial intelligence technologies in the Energy Internet, more and more cyber security risks such as high-risk vulnerabilities of industrial control devices, wireless communication risks, industrial network viruses, backdoors and advanced persistent threats (APT) are brought into the industrial control system [2]. Information flows with energy in the Energy Internet, forming a widely distributed, domain-wide data application environment. However, the openness, interconnection and sharing mechanism of the energy Internet will lead to continuous malicious network attacks, which take advantage of the coupling between cyber and

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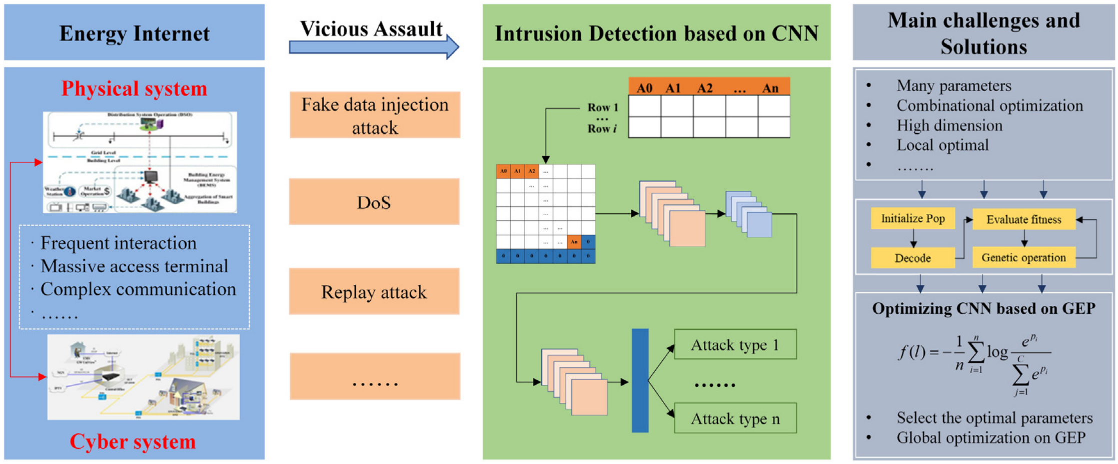

Fig. 1. Conceptual model of intrusion detection based on GEP-CNN for Energy Internet.

physical systems in the energy Internet to generate cross-space, cross-system and cross-platform chain reactions. This will pose a great threat to the security of the production, transmission, transaction and consumption business systems in the energy Internet. The complex network environment, the large number of intelligent devices and the frequent information interaction in the energy Internet make it undoubtedly to make the network attacks for the energy Internet constantly. Then how to efficiently and accurately detect intrusion attacks is crucial to the security protection of the energy Internet.

Machine learning-based intrusion detection algorithms have been widely studied in order to improve the accuracy of intrusion detection and the learning ability of unknown attacks [3,4]. However, the network traffic and features continue to increase in the open and shared energy Internet environment, leading to the increasing computational cost of existing machine learning-based intrusion detection algorithms. Meanwhile, the dimensionality reduction of attack features by existing algorithms may corrupt or eliminate the implicit correlation between intrusion data and miss a part of valid features or feature combinations.

In recent years, many scholars have proposed intrusion detection algorithms based on convolutional neural networks (CNN) [5,6]. These algorithms utilized the feature learning capability of CNN to abstract high-dimensional intrusion features into low-dimensional intrusion features through nonlinear matrix transformations to achieve optimization of intrusion features, preserve strongly correlated features for attack classification, and improve the accuracy of attack detection. In addition to CNN, many other AI methods have been proposed [7–10]. However, these methods, like CNNs, suffer from complex structures, numerous parameter combinations, and tends to fall into local optimality. Sun et al. [11] proposed a GA-based convolutional neural network optimization algorithm and applied it to image classification. Suganuma et al. [12] used Cartesian Genetic Programming (CGP) to optimize the structure of CNNs to improve the computational efficiency and accuracy of computer vision tasks. Compared to Genetic Algorithms (GA) and Genetic Programming (GP), Gene Expression Programming (GEP) has powerful global search and optimization capabilities [13–15]. It combines the advantages of GA and GP to solve complex problems based on simple coding. To improve the adaptability of parameter selection in CNN and avoid falling into local optimum, this paper proposes the intrusion detection algorithm based on GEP-CNN (GCNN-IDS), which achieves the global optimization of CNN parameters through GEP to improve the accuracy of network intrusion detection in the energy Internet. The idea of the intrusion detection algorithm based on GEP-CNN for energy Internet is shown in Fig. 1.

In general, the main work of this paper includes:

- To process intrusion datasets containing high-dimensional features without destroying the implicit correlations between intrusion features, this paper proposes an intrusion detection algorithm based on CNN (CNN-IDS). The local correlation of features is accurately extracted using CNN to improve the accuracy of intrusion feature extraction, thus further improving the accuracy and precision of intrusion detection.
- To address the problems of complex structure, numerous parameter combinations and easy to fall into local optimality in CNN, the intrusion detection algorithm based on GEP-CNN (GCNN-IDS) is proposed by combining the powerful global search capability of GEP. In GCNN-IDS, we design a new GEP coding and fitness function to optimize the number of convolutional and fully connected layers of CNN.
- Comparative experiments are conducted on benchmark datasets and grid-oriented real attack dataset. The experimental results demonstrate that the intrusion detection model proposed in this paper has higher intrusion detection performance compared to neural network (ANN), logistic regression (LR) and Naive Bayesian algorithm (NB).

The remainder of this paper is organized as follows. The related work is focused on in Section 2. We analyzed intrusion detection algorithm based on CNN in Section 3. Section 4 gives an intrusion detection algorithm based on GEP-CNN. Experiment results and analyses on benchmark and real dataset for power system are shown in Section 5. Finally, Conclusion and future work are described in Section 6.

2. Related work

In this section, we review network attack analysis and attack detection for power CPS, and discuss the pros and cons of these models and their relationship with our proposed model.

2.1. Network attack analysis for power cyber–physical system

Network security protection for power cyber–physical system (CPS) is crucial to ensure the safe and stable operation of power systems. The accurate analysis of various types of network attacks is an important prerequisite for intrusion detection and has introduced extensive attention from many researchers.

Ding et al. [16] proposed a distributed recursive filtering algorithm for CPS to solve spoofing attacks in CPS and verified the effectiveness of the algorithm on IEEE 39-Bus system. Sahu et al. [17] designed a testbed for CPS and used man-in-the-middle and denial-of-service attacks on it to demonstrate false data attacks and injection attacks, and validated them on a large-scale synthetic grid. Hossain et al. [18] pointed out that the wide application of information and communication technology made the smart grid vulnerable to cyber attacks, and analyzed the types of attacks and detection methods for cyber and physical systems in the smart grid. Su et al. [19] discussed the impact of cyber attacks on the state estimation of power CPS and how the attack process can be reconstructed by linear matrix inequalities. Tian et al. [20] proposed an algorithm to prevent false data injection attacks from affecting the state estimation of power CPS, which incorporates protection strategies against two types of false data injection attacks with CPS-related parameters. Zhang et al. [21] summarized the existing work related to network attacks under CPS and proposed a process for detecting CPS network attacks based on deep learning. Tu et al. [22] analyzed the consequences of cyber attacks on CPS by fusing integrity attacks and availability attacks and applying them to traditional state estimation. Hu et al. [23] designed an cyber physical moving target defense (CPMTD) strategy to prevent and detect cyber attacks against power cyber physical systems. The effectiveness of the CPMTD strategy against false data injection attacks was also verified on IEEE 14-Bus and 57-Bus. Wlazlo et al. [24] analyzed man-in-the-middle (MiTM) attacks that could exploit seemingly normal information flows to perform false command injection attacks, false data injection attacks, and replay attacks on smart grids. To detect false data injection attacks against power CPS, Wu et al. [25] proposed a state reconstruction algorithm based on an extreme learning machine and verified the effectiveness and efficiency of the proposed algorithm on several IEEE Bus systems. Wu et al. [26] designed two types of false data injection attack models based on partial feedback of generator frequencies in power CPS and verified the effectiveness of the proposed models in IEEE 9-Bus system.

2.2. State-of-the-art intrusion detection algorithms

In the above-mentioned literatures, researchers have analyzed security threats and attack protection methods for energy information physical systems. How to accurately detect these attacks is crucial for the security protection of energy information physical systems. This subsection focuses on various state-of-the-art intrusion detection algorithms.

Intrusion detection technologies can be divided into two main categories: Signature-based intrusion detection systems (SIDS) and Anomaly-based intrusion detection system (AIDS). Among them, SIDS uses pattern matching techniques to find known attacks. To improve the performance of SIDS, Felix et al. [27] proposed an intrusion detection system that fused IPFIX-based flow monitoring with NIDS, which improved the detection performance with the same detection rate. Wang et al. [28] proposed a fog computing-based privacy-preserving approach to address the privacy protection of shared data in SIDS. Li et al. [29] gave a blockchain-based approach for verifying shared signatures, which enhanced the robustness and effectiveness of SIDS.

SIDS is designed to detect attacks on known signatures. For new types of attacks, the detection accuracy of SIDS is greatly reduced. And, AIDS can effectively solve this problem. Xiao et al. [30] proposed a machine learning-based anomaly intrusion detection technique considering privacy-preserving scenarios. Khraisat et al. [31] proposed anomaly detection algorithm based on decision tree classification to minimize the false alarm rate of intrusion detection. Shen et al. [32] proposed a hybrid-augmented device fingerprinting method to improve the traditional intrusion detection mechanism. Experimental results demonstrated the better robustness and effectiveness of the proposed method against forgery attacks and intrusions.

Meanwhile, the increasing amount of network traffic poses a great challenge to intrusion detection systems, and the detection performance is closely related to the selected features and classifiers, while the above intrusion detection algorithms cannot be applied to the detection of network intrusion data streams with high-dimensional features. Deep learning can well achieve feature selection for high-dimensional data without destroying the correlation between the original data features, and has been widely used for intrusion detection [7,33,34]. Nathan et al. [35] proposed an intrusion detection algorithm based on asymmetric deep autoencoder, and the experimental results showed that the algorithm had higher accuracy compared with traditional NIDS. Riyaz et al. [36] fused CNN with feature selection algorithm based on conditional random field and linear correlation coefficient, and applied it to intrusion detection system. Li et al. [37] proposed an intrusion detection algorithm for industrial CPS based on federated deep learning, and the experimental results on real datasets demonstrated that the proposed model had high detection efficiency while privacy preserving.

Sasanka et al. [38] constructed a CNN-based multi-class intrusion detection system, and the experimental results on datasets such as NSL-KDD and UNSW-NB 15 showed that the algorithm had better intrusion detection performance. For the imbalanced attack classification problem, Azizjon et al. [39] proposed a network IDS system based on one-dimensional convolutional neural network (1D-CNN). Compared with classical machine learning classifiers, experimental results showed that 1D-CNN had good performance in network intrusion detection. Nguyen et al. [5] applied genetic algorithm, fuzzy C-means clustering and convolutional neural network to network intrusion detection system in order to improve the performance of intrusion detection. El-Sayed et al. [40] proposed a novel intrusion detection algorithm for software define network based on the convolutional neural network (CNN) and regularizer method to solve the overfitting of model. Kan et al. [41] proposed a intrusion detection architecture for IOT by using particle swarm algorithm to optimize the structure parameters of CNN. Experimental results showed that the model was effective and reliable. Zhang et al. [42] proposed an intrusion detection model based on convolutional neural network which combined with SMOTE and Gaussian mixture model. Experimental results on benchmark datasets indicated that the model outperformed the other intrusion detection algorithms.

Although CNN has been widely used in intrusion detection, CNN suffers from complex structure, numerous parameters and the tendency to fall into local optimality, thus making it very difficult for us to obtain a high-precision intrusion detection model for the energy Internet. In this paper, we propose a GEP-CNN based intrusion detection algorithm for the energy Internet by combining the powerful global search capability of gene expression programming to improve the optimal selection of CNN parameter combinations.

3. CNN-IDS

In this section, we propose an intrusion detection system based on convolutional neural network (CNN-IDS).

3.1. Convolutional neural network

convolutional neural network (CNN) is a feedforward neural network with a deep structure that includes convolutional computation, and is one of the representative algorithms of deep learning [43]. A typical CNN consists of an input layer, several convolution layers, several pool layers, several fully connected

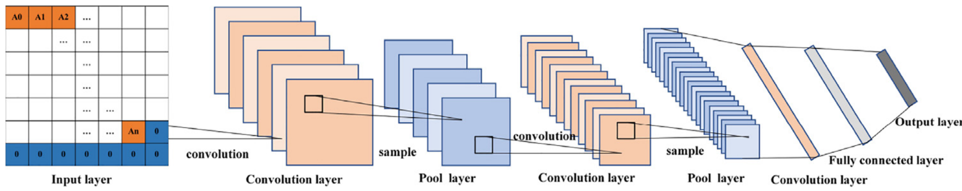

Fig. 2. CNN structure.

layers, and an output layer. Then the CNN model is evaluated by a loss function. The structure of a typical CNN model is shown in Fig. 2.

(1) Input layer

Since the gradient descent algorithm is used for learning in CNN, the input dataset needs to be normalized. If the input data are pixels, the original pixel values distributed in [0, 255] can be normalized to [0, 1]; if the input data are strings, the strings are first converted to a certain character, then to the corresponding ASCII value, and finally normalized to [0, 1]; if the input data are numeric, they are directly normalized to [0, 1]. The normalization of input data is beneficial to improve the learning efficiency and performance of CNN.

(2) Convolution layer

The essence of the convolution operation is to extract feature information from a dataset.

(3) Pool layer

The pool layer is located in the middle of the successive convolutional layers and is used to compress the number of data and parameters, reduce the dimensionality of the input features, and reduce overfitting.

(4) Fully connected layer

Fully connected layers in CNN are equivalent to the implicit layers in traditional feedforward neural networks and pass signals only to other fully connected layers. In CNN, there can be one or more fully connected layers, which are mainly used to sum the weights of the output features of the convolution and pool layers.

(5) Output layer

In general, the output layer of CNN uses a normalized exponential function (softmax function) to output the classification labels.

Let the input data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ have s categories, i.e., $y_i \in \{1, 2, \ldots, s\}$, ϑ_j,$j = \{1, 2, \ldots, s\}$ is the model parameters corresponding to the jth category. softmax function estimates the probability $p(y_i = j|x_i : \vartheta)$ that $x_i, i \in \{1, 2, \ldots, n\}$ belongs to each class by Eq. (1).

$$p(y_i = j|x_i : \vartheta) = \frac{e^{\vartheta_j x_i}}{\sum_{k=1}^{s} e^{\vartheta_k x_i}} \tag{1}$$

And $p(y_i = j|x_i : \vartheta)$ needs to satisfy both the following properties.

(1) $p(y_i = j|x_i : \vartheta) \geq 0$;
(2) $\sum_{i=1}^{n} p(y_i = j|x_i : \vartheta) = 1$.

(6) Loss function

The loss function is a measure of learning performance. The learning objective is to adjust the parameters of the CNN to minimize the loss function.

Definition 1. Let each output of the CNN input layer be x^r, where $r = 1, 2, \ldots, R$, the network weight be W^r, the offset be O^r, and we have $V^r = W^r x^{r-1} + O^r$. Then, $x^r = g(V^r)$ is called the activation function.

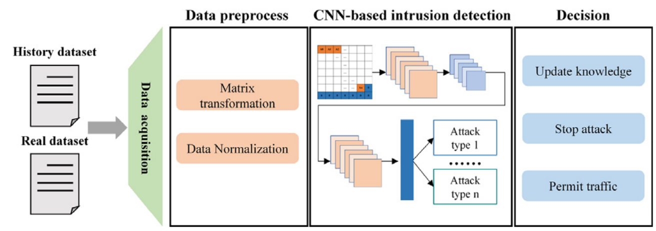

Fig. 3. Framework of CNN-IDS.

Definition 2. Let the size of the sample data be n, the number of classifications be c, the actual value of the kth class in the nth sample be u_k^n, and the output value of the kth class in the nth sample be y_k^n. Then we call $E^n = \frac{1}{2}\sum_{k=1}^{c}\left(u_k^n - y_k^n\right)^2 = \frac{1}{2}\left\|u^n - y^n\right\|_2^2$ the loss function of the CNN.

3.2. CNN-IDS description

For the requirements of intrusion detection, a highly abstract form of intrusion data can both improve the classification of intrusion attacks and reduce the influence of feature components that are not relevant to the classification of intrusion attacks on the classification results. In this paper, we propose an intrusion attack detection algorithm based on convolutional neural network (CNN-IDS). The framework of the CNN-IDS is shown in Fig. 3.

From Fig. 3, we can see that CNN-IDS mainly includes data acquisition, data preprocess, CNN-based intrusion detection model and decision making. The whole steps are shown as follows.

(1) First, on the trained intrusion detection model, the network packet capture software is used to capture the network data stream in real time and parse out the corresponding network data.

(2) Then, the input datasets such as text, string, etc. are transformed into a matrix format that satisfies the CNN input requirements by matrix transformation and data normalization.

For CNN-IDS, it is crucial to convert the intrusion dataset containing text, characters, into the matrix format required by the CNN network. The idea of intrusion data pre-processing is shown in Fig. 4.

Let the intrusion dataset $I_d = [X_1, X_2, \ldots, X_m]$, where $X_i = [x_{i1}, x_{i2}, \ldots, x_{in}], i = \{1, \ldots, m\}$. The process of intrusion data preprocessing (IDP) is shown as follows.

(3) Finally, a CNN-based intrusion detection model is constructed to achieve high accuracy detection of n different types of attacks.

The algorithm description of CNN-IDS is shown as follows.

4. GCNN-IDS

In CNN-based intrusion detection algorithms, the parameters of the CNN, especially the number of convolution and fully connected layers, have a significant impact on the performance of intrusion detection [41,42]. Generally, we construct the initial

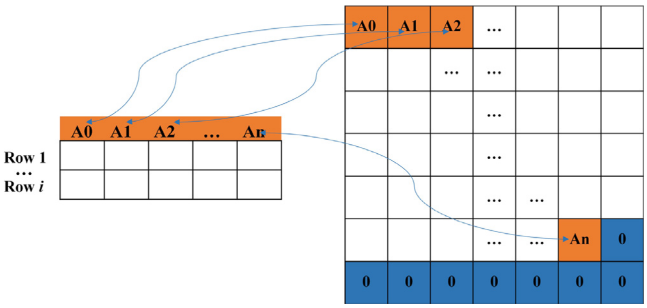

Fig. 4. Intrusion data preprocessing.

Algorithm 1: IDP

Input: $I_d = [X_1, X_2, \ldots, X_m]$;
Output: CNN dataset C_d;
begin
  for $i \leftarrow 1$ *to* m **do**
    $X_i \leftarrow ReadeachRow(I_d)$;
    if X_i *is String* **then**
      $X_i \leftarrow ConvertoChar(X_i, char)$;
      $X_i \leftarrow ConvertoAsc(X_i, ASCII)$;
      $X_i \leftarrow \frac{X_i - \min(X_i)}{\max(X_i) - \min(X_i)}$;
      $C_d \leftarrow Insert(X_i, C_d)$;
    else if X_i *is Number* **then**
      $X_i \leftarrow \frac{X_i - \min(X_i)}{\max(X_i) - \min(X_i)}$;
      $C_d \leftarrow Insert(X_i, C_d)$;
    end
  end
  end
  Return C_d;
end

Algorithm 2: CNN-IDS

Input: $I_d = [X_1, X_2, \ldots, X_m]$;
Output: Intrusion detection accuracy A_c;
begin
  $C_d \leftarrow IDP(I_d)$;
  for $i \leftarrow 1$ *to* L **do**
    $O_i \leftarrow Convolution(C_d, Kernal_i)$;
    $O_i^{'} \leftarrow Pool(O_i)$;
  end
  for $j \leftarrow 1$ *to* K **do**
    $O_j^{'} \leftarrow FullConnect(O_j^{'}, Activef_j)$;
  end
  for $r \leftarrow 1$ *to* R **do**
    $l_r \leftarrow -\frac{1}{n}\sum_{i=1}^{n} \log \frac{e^{Pr}}{\sum_{r=1}^{R} e^{Pr}}$;
  end
  $A_c \leftarrow softmax(l)$;
  Return A_c;
end

CNN by randomly initializing the number of convolution and fully connected layers, and then continuously adjust the number of convolution and fully connected layers to optimize the accuracy and precision of intrusion detection according to the results of intrusion detection. It is very difficult to automate the selection of the number of convolution and fully connected layers to achieve the optimal accuracy and precision of intrusion detection. In this paper, an intrusion detection algorithm based on GEP-CNN (GCNN-IDS) is proposed. The biggest advantage of using GEP to optimize the combination of the number of convolution and fully connected layers in CNN is that the algorithm does not need to give the initial combination of convolution and fully connected layers in advance, and the optimal selection is made entirely by the encoding, decoding and genetic operations of the GEP algorithm.

4.1. CNN parameter optimization based on GEP

4.1.1. GCNN coding

The first thing to do in the GCNN-IDS algorithm is how to encode the CNN structure. The encoding of GCNN has to satisfy both the basic structure of GEP and reflect the CNN structure. After the CNN structure is encoded correctly, the mutation and recombination operations can be performed.

Let $C_{NN} = \{C, P, F\}$ denote a convolutional neural network, where C denotes the convolution layer, P denotes the pool layer, and F denotes the fully connected layer. A complete convolutional neural network is a different combination of C, P and F. Since the experimental dataset in this paper corresponds to a small number of features, there is no need to perform additional feature reduction. Therefore, in this paper, in order to simplify the coding structure of GEP, GCNN only considers convolution layer and fully connected layer without pool layer.

To construct the encoding of GCNN, two new functions and are designed in this paper. For two arbitrary sets or elements A and B, the following properties are given.

Property 1.
(1) $A \cup B = B \cup AA \cap B = B \cap A$;
(2) $A \cup B = \{A, B\}, A \cap B = \emptyset$;
(3) $\emptyset \cup A = \{A\}, \emptyset \cap A = \emptyset$;
(4) $A \cup A = \{A, A\}, A \cap A = \{A\}$;
(5) $A \cup \{A, B\} = \{A, A, B\}, A \cap \{A, B\} = \{A\}$.

Definition 3. Let a binary set $G = \langle G_h, G_t \rangle$, where G_h denotes the gene head and G_t denotes the gene tail. And let the function set $F = \{\cup, \cap\}$ and the terminal set $T = \{C, F\}$. G_h is generated randomly from F and T, and G_t can only be generated randomly from T. Then the binary sequence $G = \langle G_h, G_t \rangle$ is called GCNN gene.

Definition 4. Let $G_i = \langle G_h, G_t \rangle$ be any of the GCNN genes, then $R = \{G_i | i \in [1, n]\}$ is called the GCNN chromosome.

Definition 5. Let R_i be any one GCNN chromosome, then $P = \{R_i | i \in [1, Pop_{size}]\}$ is called GCNN population.

Property 2. *Let the length of G_h be l_h, the length of G_t be l_t, and the maximum number of operands of F in G_h be $\phi(F)$. Then the following equation holds.*

$$l_t = l_t \times (\phi(F) - 1) + 1 \tag{2}$$

Example 1. Let $F = \{\cup, \cap\}$, $T = \{C, F\}$, l_h=5, and the randomly generated GCNN chromosome is shown as follows.

As can be seen in Fig. 5, the GCNN chromosome includes 2 GCNN genes. We encode each GCNN gene into the corresponding expression tree and obtain the corresponding expression by decoding. According to Property 1, we get the number of convolutional and fully connected layers in the CNN as 6, 2, respectively. This coding pattern greatly reduces the chromosome structure, which makes it have faster convergence speed.

Meanwhile, to ensure the validity of the CNN structure, the following theorem holds.

Lemma 1. *If $\forall G_i \cap T = \{C\}$ or $\forall G_i \cap T = \{F\}$, then we consider G_i to be a null GCNN gene.*

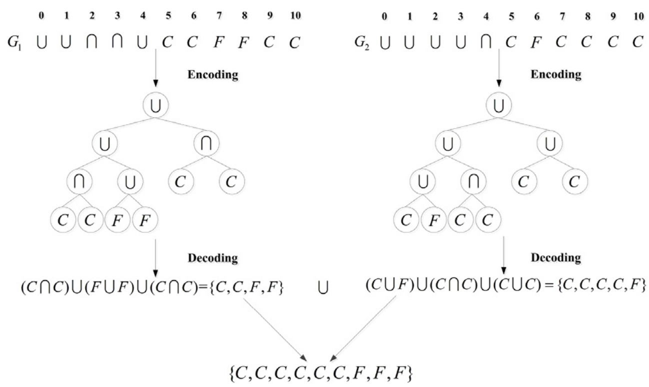

Fig. 5. Chromosome of GCNN.

Proof. Because at least one convolution layer C and one fully connected layer F must be included in the CNN. If $\forall G_i \cap T = \{C\}$ or $\forall G_i \cap T = \{F\}$ holds, it means that the convolutional neural network represented by the currently generated GCNN gene has only convolution or fully connected layers, then the CNN structure generated by the gene does not meet the requirements, so we think that the G_i is an invalid GCNN gene.

4.1.2. Fitness function

The fitness function is designed to evaluate whether the parameters corresponding to the GCNN chromosomes in the current GEP population are optimal. Usually, individuals with high fitness will be applied to genetic operations with higher probability. In the GCNN-IDS algorithm, we choose the loss function as the fitness function of GCNN. The loss function is used to estimate the degree of inconsistency between the predicted value $f(x)$ of the GCNN model and the true value y. The smaller the loss function is, the better the model is.

Definition 6. Let n be the size of the intrusion dataset, C be the number of attack types, x_i denote the ith intrusion feature, y_i be the label of the attack type corresponding to x_i, w_j and b_j be the weight and displacement of attack type j. Then, $f(l) = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{e^{p_i}}{\sum_{j=1}^{C} e^{p_i}}$ is called the loss function of the GCNN model.

In this paper, in order to objectively evaluate the performance of CNN after GEP optimizes the number of convolution and fully connected layers of CNN, we choose $f(l)$ as the fitness function of GEP.

4.2. GCNN-IDS

In GCNN-IDS, we use GEP to automatically select the optimal combination of convolution and fully connected layers, update them into CNN to train on the intrusion dataset, learn the intrusion features, and finally classify the attack types. The flow of the GCNN-IDS algorithm is shown in Fig. 6.

From Fig. 6, we understand that in the GCNN-IDS algorithm, the GEP population $P = \{R_i | i \in [1, Pop_{size}]\}$ is first initialized according to various parameters (including population size, maximum evolutionary generation, mutation probability, recombination probability, function set, terminal set, etc.); then, all chromosomes $R = \{G_i | i \in [1, n]\}$ in P are decoded and the initial number of convolution and fully connected layers $\{C_i, F_i | i \in [1, Pop_{size}]\}$ are calculated. Then, the Pop_{size} CNN networks are initialized according to $\{C_i, F_i | i \in [1, Pop_{size}]\}$, and the loss value $\{f(l_1), \ldots, f(l_{Pop_{size}})\}$ is calculated by feeding the pre-processed intrusion training dataset into the Popsize CNN network for training, respectively. If $\min(f(l_1), \ldots, f(l_{Pop_{size}})) < \delta$, the corresponding intrusion attack detection result is output directly, otherwise the mutation and recombination genetic operations in the GEP algorithm are executed to generate a new GCNN population. This illustrates that the GCNN-IDS algorithm proposed in this paper does not need to initialize the number of convolution and fully connected layers in the CNN, and controls multiple CNN networks to train intrusion data simultaneously by threading, which improves the efficiency of optimal selection of parameter combinations in the CNN, and also ensures the accuracy of intrusion detection.

The algorithm description of GCNN-IDS is shown as follows.

Algorithm 3: GCNN-IDS

Input: $Popsize$, $MaxGen$, p_m, p_c, F, T;
Output: Intrusion detection accuracy A_c;
begin
  $Pop \leftarrow Init(MaxGen, Popsize, p_m, p_c)$;
  $Pop \leftarrow Encode(Pop, F, T)$;
  while $gen \leq MaxGen$ **do**
    $Str \leftarrow Decode(Pop, InOrder)$;
    $C[r], F[s] \leftarrow Cout(Str)$;
    $CNN[Popsize] \leftarrow BuildCNN(C[r], F[s])$;
    for $q \leftarrow 1$ *to* $Popsize$ **do**
      $f(l_q) \leftarrow -\frac{1}{n}\sum_{i=1}^{n} \log \frac{e^{p_i}}{\sum_{j=1}^{C} e^{p_i}}$;
    end
    if $\min(f(l_1), \ldots, f(l_{Popsize})) \leq \delta$ **then**
      $A_c \leftarrow softmax(\min(f(l_1), \ldots, f(l_{Popsize})))$;
    else
      $Pop \leftarrow Mutation(Pop, p_m)$;
      $Pop \leftarrow Recommbination(Pop, p_r)$;
      $Pop \leftarrow GenerateNewPop(Pop, F, T)$;
      $Gen \leftarrow Gen + 1$;
    end
  end
  end
  Return A_c;
end

5. Experiments and analysis

5.1. Experimental setting and data description

To test the performance and effectiveness of the algorithm proposed in this paper, the test platform is Windows 10 64-bit+Inter(R) Core(TM) i5-3470 CPU@3.20 GHz+16 G RAM+jdk1.8+ eclipse2019+python3.7.

The experimental dataset consists of two classical intrusion detection datasets and a real dataset for attacks on the cyber physical system of power grid. To facilitate the computation, we preprocessed the above three public datasets accordingly to form the original dataset as shown in Table 1.

The KDDCUP99[1] has 311027 instances, including 41 condition features and 5 attack features. To simplify the calculation, 5171 instances are randomly selected as experimental data based on the proportion of each attack type in the original dataset, of which 70% are used as train data and the remaining 30% as test data.

There are 82,332 instances in the UNSW_NB15,[2] and each data includes 44 condition features and 10 attack features. To simplify the calculation, we randomly select 5061 instances from the original UNSW_ NB15 test-set.csv dataset based on the proportion of each attack type in the original dataset as experimental data, of

1 http://kdd.ics.uci.edu/databases/kddcup99/correct.gz
2 https://www.unsw.adfa.edu.au/unsw-canberra-cyber/cybersecurity/ADFA-NB15-Datasets/

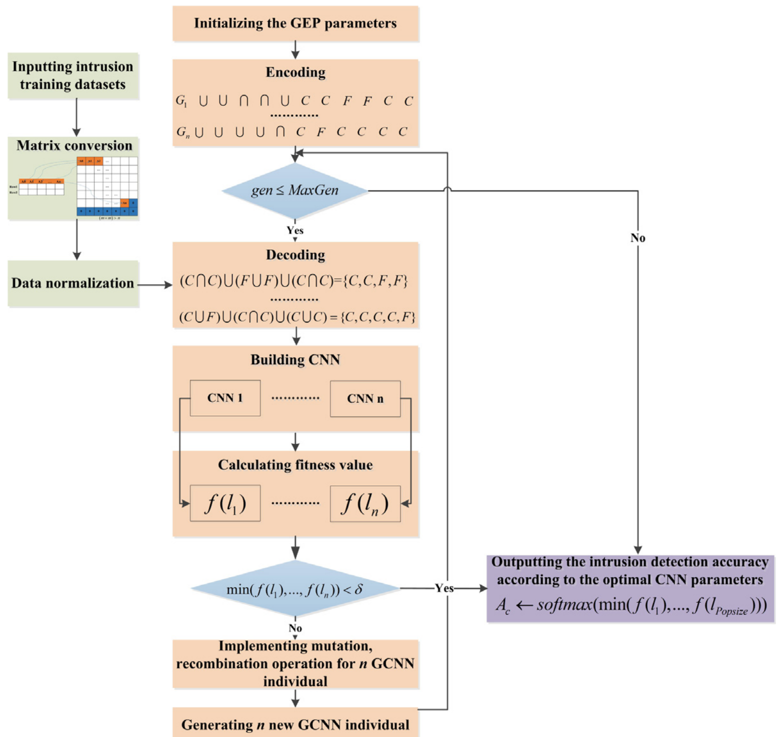

Fig. 6. The flow of GCNN-IDS.

Table 1
The experimental datasets.

Dataset		#Instance		#Condition feature	#Attack feature
KDDCUP99	Train Test	3619 1552	5171	41	5
UNSW_NB15	Train Test	3542 1519	5061	44	10
attack_data	Train Test	3302 1416	4718	40	4

which 70% are used as train data and the remaining 30% as test data.

The attack_data[3] has 85,627 instances (of which non-intrusion-related data have been removed), each including 40 condition features and 5 attack features. 4718 instances were randomly selected as experimental data, of which 70% were used as training data and the remaining 30% as test data.

5.2. Evaluation index

To evaluate the performance of the proposed algorithm in this paper, for the experimental dataset in Table 1, we use intrusion detection accuracy, precision, recall, F_1-Measure and false alarm rate to evaluate the performance of various types of intrusion detection models.

Let *TP* denote the number of normal traffic identified as normal traffic, *FP* denote the number of attack traffic identified as normal traffic, *TN* denote the number of attack traffic identified as attack traffic, and *FN* denote the number of normal traffic identified as attack traffic. Then we have the following equation.

[3] https://github.com/TouwaErioH/Machine-Learning/tree/master/classifier/pro1

Let $A_c = \frac{TP+TN}{TP+TN+FP+FN}$ denote the accuracy rate of the intrusion detection algorithm. The higher the A_c, the more effective the intrusion detection algorithm is.

Let $P_r = \frac{TP}{TP+FP}$ denote the precision of the intrusion detection algorithm. The higher the P_r, the better the intrusion detection algorithm.

Let $R_e = \frac{TP}{TP+FN}$ denote the recall rate of the intrusion detection algorithm. The higher the R_e, the better the intrusion detection algorithm.

Let $F_1 = \frac{2P_rR_e}{P_r+R_e}$ denote the F_1-measure of the intrusion detection algorithm. The higher the F_1, the better the intrusion detection algorithm.

Let $F_r = \frac{FN}{TP+FN}$ denote the false detection rate of the intrusion detection algorithm. The smaller F_r is, the better the intrusion detection algorithm is.

5.3. Experimental analysis

In this subsection, we do two experiments to verify the effectiveness and reliability of the algorithm proposed in this paper. For all train datasets in Table 1, Experiment 1 compares the accuracy of CNN-IDS by comparing different parameter combinations. Experiment 2 compares the detection accuracy, precision, recall, F_1 and false detection rate of GCNN, CNN based on GP(GPCNN) [44], CNN, artificial neural network (ANN) [45], logistic regression (LR) [46] and Naive Bayesian algorithm (NB) [47].

5.3.1. Comparison of the accuracy of CNN-IDS under different parameters

To verify the effect of different parameter settings on the performance of CNN-IDS, the experimental parameter settings of CNN are shown in Table 2.

For the KDDCUP99 train dataset in Table 1, Fig. 7 shows the comparison of the average loss values of three CNN-IDS models for different number of iterations. For the KDDCUP99 test dataset in Table 1, Fig. 8 gives a comparison of the intrusion

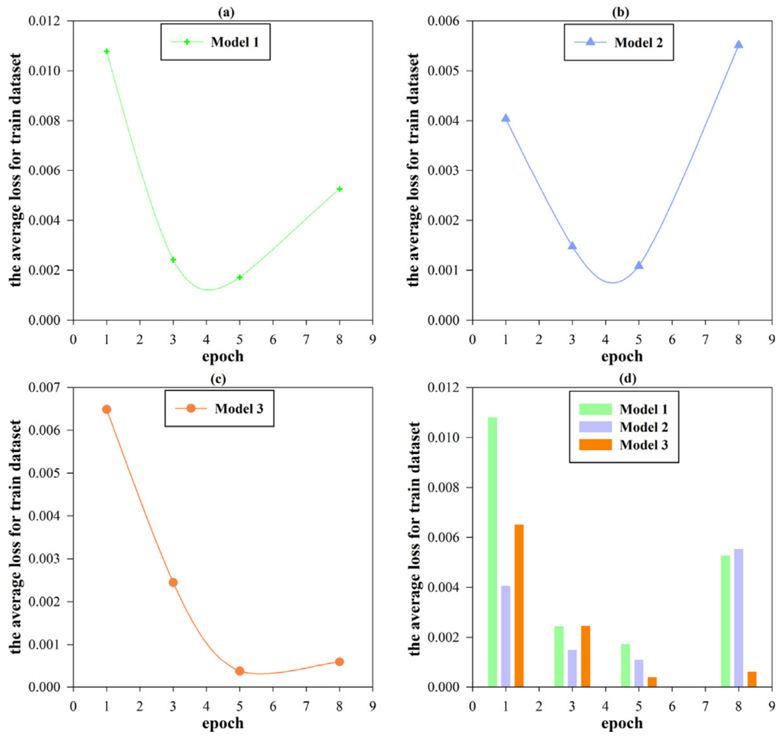

Fig. 7. Comparison of the average loss of the CNN-IDS model under different iterations.

Table 2
Parameters setting of CNN model.

Model	Convolution layer	Kernel size	Full connected layer	Activation function
Model 1	1	3 × 3 (128)	3	*Relu*
Model 2	2	3 × 3 (128)	3	*Relu*
Model 3	3	3 × 3 (128)	3	*Relu*

Table 3
Parameters setting of CNN model.

Parameter	Value	Parameter	Value
Popsize	50	GeneHead	5
MaxGen	10	Number of Genes	2
Function set	$\{\cup, \cap\}$	Link function	$\cup$
Terminal set	$\{C, F\}$	Running time	10
p_c	0.3	Batch_size	128
p_m	0.01	Active function	Relu

detection accuracy of three CNN-IDS models for different number of iterations.

From Fig. 7, we can see that for the KDDCUP99 train dataset, the average loss of the three CNN models in Table 2 first decreases and then increases as epoch increases. Compared with Model 1 and Model 2, the average loss of Model 3 increases slowly with the increase of epoch. And, when epoch equals 5, the average losses of the three models reach the minimum values of 0.00171, 0.0011, and 0.0004, respectively.

As can be seen in Fig. 8, for the KDDCUP test dataset, the detection accuracy of Model 1 and Model 2 showed a decreasing trend as epoch increased, while the detection accuracy of Model 3 increased by about 1.54%, 2.16%, and 0.76%, respectively. When epoch = 8, the detection accuracy of Model 1 and Model 3 reached the maximum value of 0.9092 and 0.9143, respectively. The experimental results in Fig. 7 and Fig. 8 show that the number of convolutional layers has a direct impact on the accuracy of intrusion detection under the same conditions of other parameters.

5.3.2. Comparison of different intrusion detection models

First, for all the train datasets in Table 1, Fig. 9 shows the variation of the fitness values of the GCNN algorithm, and the parameters of GCNN are shown in Table 3. And, Fig. 10 compares the average time-consuming between intrusion detection based on GCNN and intrusion detection based on GPCNN. Then, for all the experimental datasets, the intrusion detection models of GCNN and convolutional neural network (CNN), artificial neural network (ANN), logistic regression (LR), and Naive Bayesian algorithm (NB) are compared in terms of accuracy, precision, recall, F_1 and false detection rate. The experimental results are shown in Tables 4, 5, 6, 7 and 8, respectively.

From Fig. 9, we can see that the optimal loss value of GCNN gradually tends to 0 as epoch increases for all train datasets in Table 1. The results show that the parameters obtained from GEP optimized CNN can also make the convolutional neural network converge. The optimal individual generated by the GCNN is shown in the following equation. From Eq. (3), it can be seen that the GCNN includes three convolutional layers and three fully connected layers.

From Fig. 10, it can be seen that for all experimental datasets, the average elapsed time of GCNN-based intrusion detection is about 1.67%, 1.34%, and 1.91% lower compared to GPCNN, respectively. This is because although both GEP and GP are encoded using a nonlinear tree structure, the encoding of GEP is fixed-length while the encoding of GP is non-fixed-length. This will lead to the fact that the search space of GEP is relatively smaller than that of GP when solving complex problems such as high-dimensional intrusion detection, which in turn makes the average time-consuming by GCNN-based intrusion detection algorithms

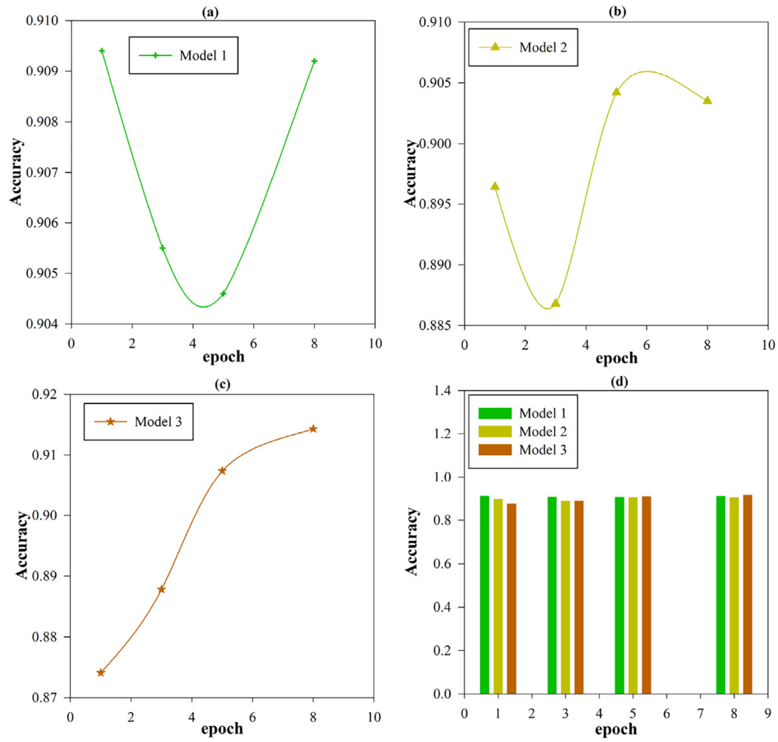

Fig. 8. Comparison of the accuracy of the CNN-IDS model under different iterations.

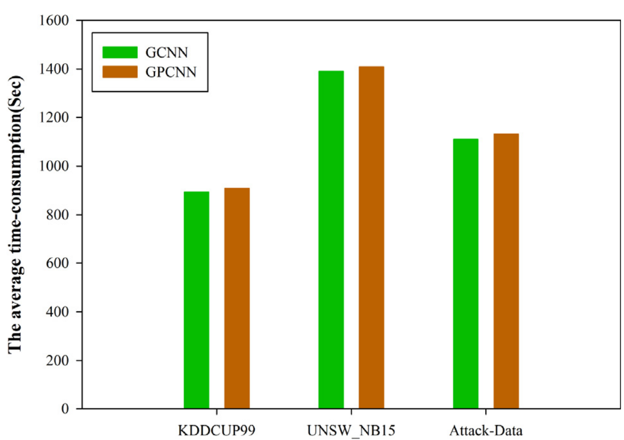

Fig. 9. The optimal loss value of the GCNN.

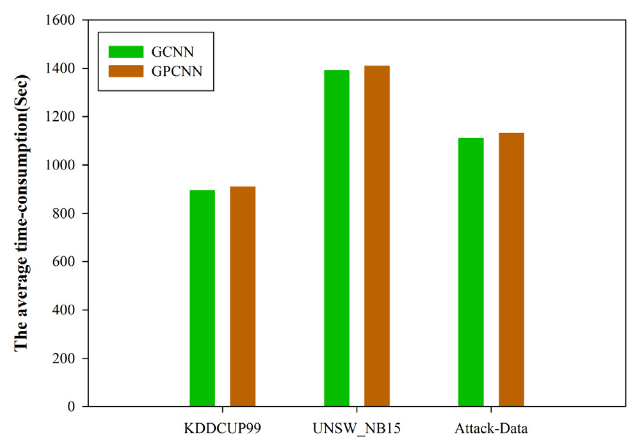

Fig. 10. Comparison of time-consuming between GCNN and GPCNN.

smaller than that of GPCNN-based intrusion detection algorithms.

$$\cup\cap\cup\cup\cup CCCFFC\cup\cup\cap F\cup FCCFCF \Rightarrow \{CCF\}\cup\{FFC\}=\{CCCFFF\} \quad (3)$$

From Table 4 to Table 8, it can be seen that for all the train and test datasets in Table 1, GCNN has the best accuracy, recall, F_1 and false detection rate compared to CNN, ANN, LR, and NB. And, the accuracy, precision, recall, F_1 and false detection rate of GCNN are slightly higher than that of GPCNN. This is because GEP is richer in genetic operations than GP, which, to some extent, makes the probability of getting optimal results based on GEP greater than GP. Specifically, the experimental results in Table 4 show that the accuracy of GCNN is maximally improved by about 56.31%, 24.5%; 74.22%, 73.11%; 3.68%, and 5.41% compared to GPCNN, CNN, ANN, LR, and NB for all train and test datasets in Table 1, respectively.

From Table 5, we know that for the attack_data train and test dataset, GCNN has the highest accuracy compared to CNN, ANN, LR and NB. For the KDDCUP99 and UNSW_NB15 train and test datasets, GCNN outperforms CNN, ANN, and LR, but is worse than NB. From Table 6, we can see that for all the train and test datasets in Table 1, GCNN has a significant advantage in recall compared to NB. And compared with CNN, ANN, and LR, the recall of GCNN is maximally improved by about 3.06%, 13.27%; 3.23%, 3.23%; 2.08%, 3.12%, respectively. The experimental results in Table 7 combine the accuracy and recall of intrusion detection, and the GCNN is

Table 4
Comparison of accuracy among various algorithms.

Dataset		GCNN	GPCNN	CNN	ANN	LR	NB
KDDCUP99	Train	**0.987**	0.979	0.96	0.96	0.92	0.63
	Test	**0.965**	0.961	0.96	0.94	0.92	0.78
UNSW_NB15	Train	**0.671**	0.668	0.64	0.64	0.53	0.39
	Test	**0.667**	0.661	0.64	0.64	0.61	0.39
attack_data	Train	**0.99**	0.98	0.98	0.99	0.95	0.98
	Test	**0.979**	0.977	0.977	0.97	0.93	0.76

Table 5
Comparison of precision among various algorithms.

Dataset		GCNN	GPCNN	CNN	ANN	LR	NB
KDDCUP99	Train	0.96	0.96	0.9	0.87	0.73	0.98
	Test	0.94	0.94	0.89	0.84	0.79	0.96
UNSW_NB15	Train	0.78	0.77	0.7	0.63	0.66	0.9
	Test	0.76	0.76	0.69	0.63	0.65	0.87
attack_data	Train	**0.99**	0.98	0.99	0.99	0.99	0.9
	Test	**0.98**	0.98	0.97	0.98	0.96	0.9

Table 6
Comparison of recall among various algorithms.

Dataset		GCNN	GPCNN	CNN	ANN	LR	NB
KDDCUP99	Train	**0.98**	0.98	0.95	0.96	0.97	0.66
	Test	**0.98**	0.97	0.94	0.93	0.85	0.44
UNSW_NB15	Train	**0.93**	0.93	0.9	0.91	0.91	0.34
	Test	**0.93**	0.93	0.9	0.90	0.91	0.34
attack_data	Train	**0.96**	0.96	0.94	0.96	0.94	0.6
	Test	**0.96**	0.95	0.93	0.96	0.94	0.6

Table 7
Comparison of F_1 among various algorithms.

Dataset		GCNN	GPCNN	CNN	ANN	LR	NB
KDDCUP99	Train	**0.97**	0.97	0.92	0.91	0.83	0.79
	Test	**0.96**	0.96	0.89	0.88	0.82	0.6
UNSW_NB15	Train	**0.85**	0.84	0.78	0.74	0.77	0.49
	Test	**0.84**	0.84	0.77	0.74	0.76	0.49
attack_data	Train	**0.97**	0.97	0.96	0.97	0.96	0.72
	Test	**0.97**	0.96	0.95	0.97	0.95	0.72

Table 8
Comparison of false detection rate among various algorithms.

Dataset		GCNN	GPCNN	CNN	ANN	LR	NB
KDDCUP99	Train	**0.013**	0.013	0.013	0.019	0.03	0.34
	Test	**0.015**	0.016	0.017	0.02	0.04	0.34
UNSW_NB15	Train	**0.034**	0.034	0.039	0.05	0.04	0.66
	Test	**0.04**	0.04	0.062	0.07	0.09	0.66
attack_data	Train	**0.057**	0.06	0.067	0.066	0.11	0.04
	Test	**0.017**	0.017	0.017	0.02	0.024	0.02

optimal compared to CNN, ANN, LR, and NB for all the train and test datasets. Overall, for the KDDCUP99 and UNSW_NB15 train and test datasets, although the precision of GCNN is worse than that of NB, the combined precision and recall of GCNN is better than that of NB. This also indicates that the GCNN algorithm has good intrusion detection performance on all experimental datasets.

Also, from Table 8, we can find that for all the train and test datasets, GCNN has the lowest false detection rate compared to CNN, ANN, LR, and NB. This also indicates that the GCNN algorithm has good attack detection capability.

6. Conclusion

To efficiently and accurately detect intrusion attacks introduced by the open and shared operational characteristics of the energy Internet, we propose an intrusion detection algorithm based on GEP-CNN. The main contributions include: (1) To eliminate the impact on intrusion detection performance caused by the large number of parameters and the tendency to fall into local optima due to the complex structure in CNN, we design new GEP encoding and fitness function to optimize the combination of the number of convolutional and fully connected layers of CNN, and finally construct an efficient intrusion detection algorithm based on GEP-CNN; (2) The results of comparative experiments on benchmark dataset and real dataset demonstrate that compared with existing algorithms, our proposed algorithm has higher intrusion detection accuracy, precision, recall and F_1 with lower false detection rate.

And, there are various forms of parameters and their combinations in convolutional neural networks, and different combinations of parameters have a direct impact on the performance of intrusion detection. In the future, we introduce more CNN parameters into the gene coding of gene expression programming, and use GEP to automatically select the most optimal CNN parameter combinations without affecting the evolutionary efficiency.

CRediT authorship contribution statement

Song Deng: Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition. **Xinya Yuan:** Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition. **Qianliang Li:** Data curation, Writing – review & editing. **Jie Zhang:** Data curation, Writing – review & editing. **Mengfei Sun:** Resources, Investigation. **Xiong Fu:** Investigation, Project administration, Supervision. **Lechan Yang:** Conceptualization, Methodology, Software, Writing – original draft, Formal analysis, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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